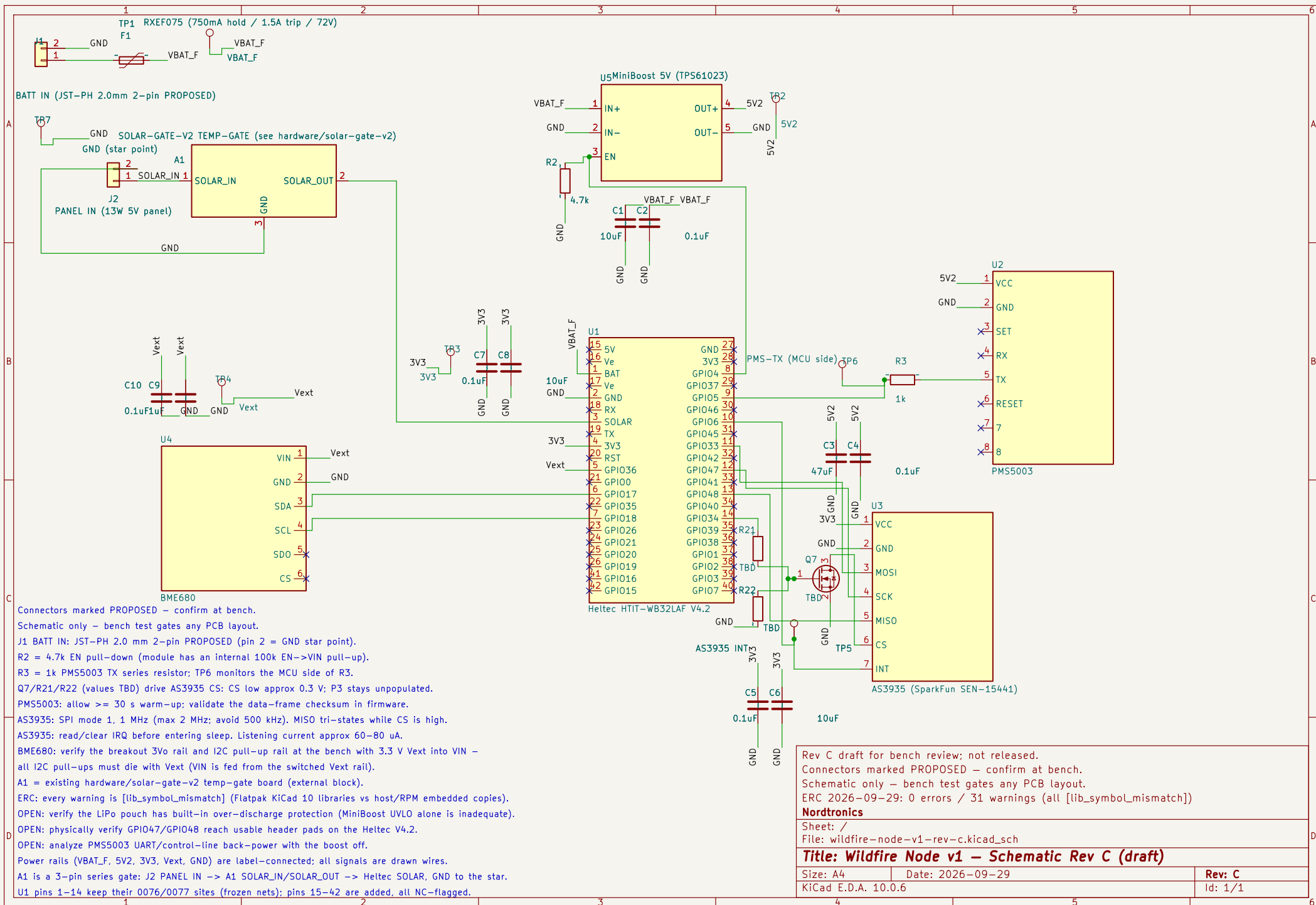




Connectors marked PROPOSED – confirm at bench.
Schematic only – bench test gates any PCB layout.
J1 BATT IN: JST-PH 2.0 mm 2-pin PROPOSED (pin 2 = GND star point).
R2 = 4.7k EN pull-down (module has an internal 100k EN->VIN pull-up).
R3 = 1k PMS5003 TX series resistor; TP6 monitors the MCU side of R3.
Q7/R21/R22 (values TBD) drive AS3935 CS: CS low approx 0.3 V; P3 stays unpopulated.
PMS5003: allow >= 30 s warm-up; validate the data-frame checksum in firmware.
AS3935: SPI mode 1, 1 MHz (max 2 MHz; avoid 500 kHz). MISO tri-states while CS is high.
AS3935: read/clear IRQ before entering sleep. Listening current approx 60–80 uA.
BME680: verify the breakout 3Vo rail and I2C pull-up rail at the bench with 3.3 V Vext into VIN – all I2C pull-ups must die with Vext (VIN is fed from the switched Vext rail).
A1 = existing hardware/solar-gate-v2 temp-gate board (external block).
ERC: every warning is [lib_symbol_mismatch] (Flatpak KiCad 10 libraries vs host/RPM embedded copies).
OPEN: verify the LiPo pouch has built-in over-discharge protection (MiniBoost UVLO alone is inadequate).
OPEN: physically verify GPIO47/GPIO48 reach usable header pads on the Heltec V4.2.
OPEN: analyze PMS5003 UART/control-line back-power with the boost off.
Power rails (VBAT_F, 5V2, 3V3, Vext, GND) are label-connected; all signals are drawn wires.
A1 is a 3-pin series gate; J2 PANEL IN -> A1 SOLAR_IN/SOLAR_OUT -> Heltec SOLAR, GND to the star.
U1 pins 1–14 keep their 0076/0077 sites (frozen nets); pins 15–42 are added, all NC-flagged.

Rev C draft for bench review; not released.
Connectors marked PROPOSED – confirm at bench.
Schematic only – bench test gates any PCB layout.
ERC 2026-09-29: 0 errors / 31 warnings (all [lib_symbol_mismatch])
Nordtronics

Sheet: /
File: wildfire-node-v1-rev-c.kicad_sch

Title: Wildfire Node v1 – Schematic Rev C (draft)

Size: A4
KiCad E.D.A. 10.0.6

Date: 2026-09-29

Rev: C

Id: 1/1